

VLSI design engineer after graduation. Most of the people who enter this industry have an engineering degree with an electronics specialisation. There is a small need for people with computer science and other specialisations as well," says Raghuram Tupuri, General Manager—Microprocessor Solutions, AMD India Engineering Center.

Chip design is a highly specialised and challenging field that involves working with sophisticated design tools, chip architectures and test methods. "Aspirants are expected to have an immense interest in and passion for the field, with a good understanding of semiconductor physics. Besides this, good logical thinking capacity and the aptitude to carry on high-intensity work is expected," says Praveen Vishakantiah, senior manager, Digital Enterprise group, Intel.

Certifications

Chip design companies look for undergraduates (B. Tech.), post-graduates (M. Tech.), and/or research-level scholars with a specialisation either in electronics, microelectronics & VLSI design, electrical engineering, or computer science and instrumentation. Candidates should have opted for one Hardware Description Language—VERILOG or VHDL—in their course of study. Sound knowledge of assembly-level programming and C programming is necessary. The most preferred qualification by the industry is an M. Tech. in Microelectronics and VLSI. Graduates from different engineering domains can also undergo post-graduation training courses provided by institutes—private and government—that offer short-term courses in VLSI design.

Obviously, certifications play a major role during the recruitment screening process. Companies prefer to employ engineers with a strong electronics background and who have studied VLSI design as a subject. "An electronics



Engineering graduates who have strong fundamentals of electrical or electronics engineering make good VLSI chip designers"

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engineer will definitely have an edge over a computer science major in the recruitment procedure, simply because we look for their practical skills towards designing chips. Freshers are expected to have good fundamentals of what they have studied," says Aruna Padmanabhan, HR Director - India, Freescale Semiconductor.

Recruitment

To reach every Regional Engineering College campus is time consuming and costly, so semiconductor companies have joined hands to form a common platform to recruit design engineers.

"Initiatives like Si-Quest (in association with Campus-Connect by ISA) act as a common platform for semiconductor companies and universities. Through such programs, students get the leverage of internships, sponsored programmes and placements," says Raj Khare, MD India, Broadcom Corporation.

The companies have set a minimum cut-off percentage (40 per cent) for candidates from any institute, for the written tests and personal interviews. The technical test focuses on digital/analog design fundamentals or software—usually C, algorithms and data structures—depending on the candidate's areas of interest and the positions available. Also, a typical aptitude test is taken. Besides this, interviewers check for basic electric and logic design knowledge. Good communication skills are a must, as is the ability to work in a team.

For experienced candidates, hands-on knowledge is also tested, on the basis of the candidate's area of interest and the capacity to work on highly complex designs and architectures.

What Comes Next?

Once absorbed into the field, one will have to undergo rigorous training programmes before handling complex design architectures of chips,

Courses

Here's a list of some of the top places that offer courses in VLSI Design and allied fields, as recommended by most VLSI chip design companies.

IIT Kharagpur: Dept. of Electronics and Electrical Communication Engineering: Post-graduate course in microelectronics and VLSI design, short term course on VLSI signal processing

IIT Delhi: Dept. of Electrical Engineering: Post-graduate course in integrated electronics and circuits; an interdisciplinary M. Tech. programme. Participating departments: Dept. of Computer Science and Engg, Dept. of Electrical Engg, Centre for Applied Research in Electronics

IIT Bombay: Dept. of Electrical Engineering: Post-graduate course in microelectronics and VLSI

IIT Chennai: Dept. of Electrical Engineering: Course in microprocessors, digital and analogue systems, VLSI design

IIT Kanpur: Dept. of Electrical Engineering: Post-graduate course in VLSI system design

IIT Guwahati: Dept. of Electronics and Communication Engineering: Post-graduate course in VLSI and signal processing

IIT Madras: Dept. of Electrical Engineering: Post-graduate course in microelectronics and VLSI design

IIT Rourkee: Dept. of Electronics and Computer Engineering: Post-graduate course in semiconductor devices and VLSI

NIT Silchar: Dept. of Electronics and Telecommunication: undergraduate course in computer science and engineering

NIT Calicut: Dept. of Electronics Engineering: Post-graduate course in microelectronics and VLSI

NIT Karnataka: Dept. of Electronics and Communication Engineering: Post-graduate course in microelectronics

NIT Trichy: Dept. of Electronics and Communication Engineering: Post-graduate course in VLSI systems

BITS, Pilani: Post-graduate course in microelectronics

BIT, Mesra: Dept. of Electronics and

Communication Engineering: Post-graduate course in instrumentation and control

Other institutes that offer short-term courses in the VLSI domain:

Bitmapper (Pune): Diploma in VLSI design
CDAC Advanced Computing Training School: Diploma in VLSI

V3Logic (Bangalore): Diploma in VLSI design

Accel Ltd (Chennai): Post-graduate diploma in VLSI design

Benms Technologies (Bangalore): Part-time / Full-time course in VLSI

Calorex (Ahmedabad): Course in VLSI ASIC chip design

Vedant (Chandigarh, Lucknow): Course in front-end and back-end design including Analogue & Mixed Signal design

Sandeepani (Bangalore, Hyderabad):

Post-graduate diploma in VLSI design, corporate training, workshop and seminars, and on-site customised training